

Semester Test 1

Semestertoets 1

FSK116

15 March 2013 / 15 Maart 2013

Surname and initials *MEMO*

Van en voorletters

Student number

Studentennummer

Signature _____

Handtekening

Time 2½ hours

Tyd *2½ ure*

Max. Total

Maks. Totaal

65 Marks Max

65 *punte Maks*

Internal Examiner: Dr J M Nel
Interne Eksaminator

External Examiner
Eksterne Eksaminator Mr R Q Odendaal

- Answer all questions.
- Clearly show all calculations. Do not just write down the answer.
- Clearly state all assumptions, rules and laws used.
- Where applicable draw the free-body diagrams.
- Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks.
- Programmable calculators may NOT be used.
- No textbooks allowed.
- A formula sheet is attached.
- All test and examination regulations of the University of Pretoria are applicable.

- *Beantwoord alle vrae.*
- *Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.*
- *Stel duidelik alle aannames, reëls en wette wat u gebruik.*
- *Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).*
- *Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.*
- *Programmeerbare sakrekenaars MAG NIE gebruik word nie.*
- *Geen handboeke word toegelaat nie.*
- *'n Formuleblad is aangeheg.*
- *Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

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Semester Test 1 **FSK 116** **Semestertoets 1**

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Venue

Lokaal

Circle: Circumference $= 2\pi r =$ omtrek Area $= \pi r^2$	$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{i}$ $+ (a_z b_x - b_z a_x) \hat{j}$ $+ (a_x b_y - b_x a_y) \hat{k}$	$T_c = T_K - 273.15^\circ$
Sphere: Area $= 4\pi r^2$ Volume $= \frac{4}{3}\pi r^3$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$T_F = \frac{9}{5}T_C + 32$
Cylinder / Silinder Area $= 2(\pi r^2) + 2\pi r h$ Volume $= \pi r^2 h$	$\Delta L = L \alpha \Delta T$ $\Delta V = V \beta \Delta T$	$v_{avg} = \frac{1}{2}(v_0 + v)$ $v = v_0 + at$
Trapezium: Area $= \frac{1}{2}(a+b)h$ $\sin^2 \theta + \cos^2 \theta = 1$ $\sin 2\theta = 2 \sin \theta \cos \theta$ $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$ $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$ $\sin \alpha \pm \sin \beta = 2 \sin \frac{1}{2}(\alpha \pm \beta) \cos \frac{1}{2}(\alpha \mp \beta)$	$Q = C(T_f - T_i)$ $Q = cm(T_f - T_i)$ $Q = Lm$ $W = \int_{V_i}^{V_f} p dV$	$x = x_0 + v_0 t + \frac{1}{2}at^2$ $x = x_0 + v_{avg} t$ $v^2 = v_o^2 + 2a(x - x_0)$ $x - x_0 = \frac{1}{2}(v_0 + v)t$ $x - x_0 = vt - \frac{1}{2}at^2$
$\vec{a} \cdot \vec{b} = ab \cos \phi$ $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$		

Question 1**6**

Convert the following to SI units:

(a) 1.00 day

[3]

1.00 day	$1.00 \text{ day} = \frac{24 \text{ hours}}{1 \text{ day}} \times \frac{60 \text{ min}}{1 \text{ hour}} \times \frac{60 \text{ s}}{1 \text{ min}} \checkmark \checkmark$ $= 86400 \text{ s} = 8.64 \times 10^4 \text{ s}$
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(b) 80 km/h

[3]

80 km/h	$80 \text{ km/h} = \frac{80000 \text{ m}}{\text{h}} \times \frac{1 \text{ h}}{60 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} \checkmark \checkmark$ $= 22.2 \text{ m/s}$
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IT IS ESSENTIAL THAT YOU INCLUDE YOUR UNITS IN YOUR MULTIPLICATION. YOU CANNOT JUST MULTIPLY NUMBERS.

Question 2

7

Vector $\vec{A}$ is directed along the $+x$ axis and vector $\vec{B}$ has a magnitude of 6.0 m. The sum of these two vectors is a third vector that is directed along the $+y$ axis, with a magnitude 3.0 times that of $\vec{A}$. What is that magnitude of $\vec{A}$?

Vektor $\vec{A}$ is langs die $+x$ -as gerig en vektor $\vec{B}$ het 'n grootte van 6.0 m. Die som van die twee vektore is 'n derde vektor wat langs die $+y$ -as gerig is met 'n grootte van 3.0 keer dié van $\vec{A}$. Wat is die grootte van $\vec{A}$?

[7]

Given information: $\vec{A} = a_x \hat{i}$ $\vec{B} = b_x \hat{i} + b_y \hat{j}$ $ \vec{B} = 6.0 \text{ m} = \sqrt{b_x^2 + b_y^2}$ $\vec{C} = \vec{A} + \vec{B} = (a_x + b_x) \hat{i} + b_y \hat{j}$ $= b_y \hat{j}$ $ \vec{A} + \vec{B} = 3.0 \vec{A} $	
Therefore $a_x = -b_x$ and $b_y = 3a_x$ $ \vec{C} = \sqrt{b_y^2} = 3.0 (\sqrt{a_x^2})$ therefore : $ b_y = 3 a_x = 3 b_x $ $ \vec{B} = 6.0 \text{ m} = \sqrt{b_x^2 + b_y^2}$ $= \sqrt{b_x^2 + (3b_x)^2} = \sqrt{10b_x^2}$ $b_x^2 = \frac{36}{10} = 3.6$ $ b_x = 1.90 \text{ m} = a_x $ $ \vec{A} = 1.9 \text{ m}$	✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓

Question 3

7

In the product $\vec{F} = q\vec{v} \times \vec{B}$, take $q = 3.0 \text{ C}$, $\vec{v} = 2.0\hat{i} + 4.0\hat{j} + 6.0\hat{k} \text{ m.s}^{-1}$ and $\vec{F} = 4.0\hat{i} + -2.0\hat{j} + 12\hat{k} \text{ N}$.

Determine $\vec{B}$ (in tesla (T)) in unit vector notation if $B_x = B_y$.

In die produk $\vec{F} = q\vec{v} \times \vec{B}$, aanvaar $q = 3.0 \text{ C}$, $\vec{v} = 2.0\hat{i} + 4.0\hat{j} + 6.0\hat{k} \text{ m.s}^{-1}$ en

$\vec{F} = 4.0\hat{i} + -2.0\hat{j} + 12\hat{k} \text{ N}$. Bepaal $\vec{B}$ (in tesla (T)) in eenheid-vektor notasie as $B_x = B_y$. [7]

$\vec{F} = q\vec{v} \times \vec{B} q(v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$ $= q((v_y B_z - v_z B_y)\hat{i} + (v_z B_x - v_x B_z)\hat{j} + (v_x B_y - v_y B_x)\hat{k})$ $= 4.0\hat{i} + -2.0\hat{j} + 12\hat{k}$ Therefore: $F_x = q(v_y B_z - v_z B_y) = 4.0$ $F_y = q(v_z B_x - v_x B_z) = -2.0$ $F_z = q(v_x B_y - v_y B_x) = 12$ If $B_x = B_y$, then $F_z = 3.0(2.0B_x - 4.0B_x) = 12\text{N}$ $-6.0B_x = 12$ $B_x = -2.0\text{T} = B_y$ And due to an error in the question, the answer was marked depending of whether you used F_y or F_x to solve for B_z. Final answer had to be in unit vector notation: $\vec{B} = -2.0\hat{i} - 2.0\hat{j} - (B_z \text{value})\hat{k}$	✓✓✓ ✓ ✓ ✓ ✓
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You need to multiply the cross product out so that you have the three components of the force vector in terms of the magnetic field components.

Question 4**6**

An aluminium-alloy rod has a length of 10.000 cm at 20.000 °C and a length of 10.020 cm at the boiling point of water.

'n Aluminium-allooi staaf het 'n lengte van 10.000 cm by 20.000 °C en 'n lengte van 10.020 cm by die kookpunt van water.

(a) What is the length of the rod at 50.000 °C?

Wat is die lengte van die staaf by 50.000 °C?

[3]

Given: $\Delta T = 80.000\text{ }^{\circ}\text{C}$ $L_o = 10.000\text{ cm}$ $\Delta L = 0.020\text{ cm}$ From the definition of expansion: $\Delta L = 0.020\text{ cm}$ $= L\alpha\Delta T$ $\alpha = \frac{\Delta L}{L\Delta T} = \frac{0.020\text{ cm}}{10.000\text{ cm}(80\text{ K})}$ $= 25.000 \times 10^{-6} / \text{C}^{\circ}$	You need to first determine α ✓
$\Delta L_{50} = L\alpha\Delta T$ $= 10.000\text{ cm}(25.000 \times 10^{-6} / \text{C}^{\circ})(30\text{ K})$ $= 75.000 \times 10^{-4}\text{ cm}$ $L = 10.0075\text{ cm at } 50^{\circ}\text{C}$	✓ ✓

(b) What is the temperature of the rod if its length is 10.003 cm?

Wat is die temperatuur van die staaf indien die lengte 10.003 cm is?

[3]

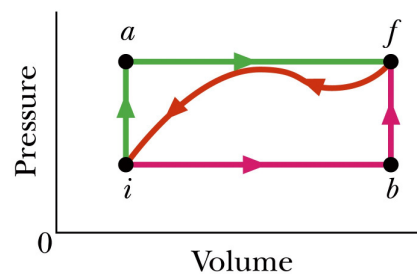
Given: $\Delta T = 80.000\text{ }^{\circ}\text{C}$ $L_o = 10.000\text{ cm}$ $\Delta L = 0.003\text{ cm}$ $\Delta L = 0.003\text{ cm}$ $= L\alpha\Delta T$ $\Delta T = \frac{\Delta L}{L\alpha} = \frac{0.003\text{ cm}}{10.000\text{ cm}(25.000 \times 10^{-6} / \text{C}^{\circ})}$ $= 12^{\circ}\text{C}$ $T = 32^{\circ}\text{C}$	✓✓ please use α from part (a) ✓ ✓
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Question 5

When a system is taken from state i to state f along path iaf in the figure shown, $Q = 210 \text{ J}$ and

$W = 84 \text{ J}$. Along path ibf $Q = 150 \text{ J}$.

Wanneer 'n stelsel van toestand i na toestand f geneem word langs die pad iaf soos aangedui in die figuur is $Q = 210 \text{ J}$ en $W = 84 \text{ J}$. Langs pad ibf is $Q = 150 \text{ J}$.



[3]

(a) Determine W along path ibf .

Bepaal W langs pad ibf .

For path iaf :	
$\Delta E_{\text{int}} = Q - W = 210 \text{ J} - 84 \text{ J} = 126 \text{ J}$	✓
This is the same for all paths $i-f$. For path ibf :	
$\Delta E_{\text{int}} = Q - W$	✓
$W = Q - \Delta E_{\text{int}} = 150 \text{ J} - 126 \text{ J}$	✓
$= 24 \text{ J}$	

(b) If $W = -54 \text{ J}$ for the return path fi , determine Q for this path.

Indien $W = -54 \text{ J}$ vir die pad fi , bepaal Q vir hierdie pad.

[3]

For path fi :	
$\Delta E_{\text{int}} = Q - W$	
$Q = \Delta E_{\text{int}} + W = -126 \text{ J} + (-54 \text{ J})$	✓✓ negative ΔE !and W
$= -180 \text{ J}$	✓

(c) If $E_{\text{int},i} = 41 \text{ J}$, what is $E_{\text{int},f}$?

Indien $E_{\text{int},i} = 41 \text{ J}$, wat is $E_{\text{int},f}$?

[3]

From (a) and path iaf :	
$\Delta E_{\text{int}} = Q - W = 210 \text{ J} - 84 \text{ J} = 126 \text{ J}$	✓
Thus	
$\Delta E_{\text{int}} = E_{\text{int},f} - E_{\text{int},i}$	✓
$E_{\text{int},f} = \Delta E_{\text{int}} + E_{\text{int},i}$	
$= 126 \text{ J} + 41 \text{ J}$	
$= 167 \text{ J}$	✓

Question 6

When the acceleration is constant, the average and instantaneous acceleration are equal:

$$a = a_{avg} = \frac{v - v_0}{t - 0}, \text{ where } a \text{ is the acceleration, } v \text{ is the velocity at time } t \text{ and } v_0 \text{ is the initial velocity at}$$

$t = 0$ s. We can also write the average velocity as $v_{avg} = \frac{x - x_0}{t - 0}$, where x is the position at time t and x_0 is the initial position at $t = 0$ s. Using these two equations show that when a particle is accelerating in the x direction, its velocity is given by $v^2 = v_0^2 + 2a(x - x_0)$. Show all steps and state all reasoning clearly.

Wanneer die versnelling konstant is, is die gemiddelde en oombliklike versnelling dieselfde:

$$a = a_{gem} = \frac{v - v_0}{t - 0} \text{ waar } a \text{ die versnelling is, } v \text{ die snelheid by tyd } t \text{ en } v_0 \text{ die aanvanklike snelheid by } t$$

$$= 0 \text{ s is. Ons kan ook die gemiddelde snelheid as } v_{gem} = \frac{x - x_0}{t - 0} \text{ skryf, waar } x \text{ die posisie by tyd } t \text{ en } x_0$$

die aanvanklike posisie by } t = 0 \text{ s is. Deur gebruik te maak van hierdie twee vergelykings, bewys dat } v^2 = v_0^2 + 2a(x - x_0). \text{ Wys al die stappe en verduidelik jou aannames.}

[7]

$a = a_{avg} = \frac{v - v_0}{t - 0} \quad \textcircled{1}$	✓
$v = v_0 + at$	
$v_{avg} = \frac{x - x_0}{t - 0} \quad \textcircled{2}$	✓
$x = x_0 + v_{avg} t$	✓
But $v_{avg} = \frac{1}{2}(v + v_0)$.	✓
Substituting this for v_{avg} in $\textcircled{2}$:	
$x = x_0 + \left[\frac{1}{2}(v + v_0) \right] t$	✓
From the first equation, we substitute for v and rearrange the terms:	
$x = x_0 + \left[\frac{1}{2}(v + v_0) \right] t$	✓
$x - x_0 = \frac{1}{2}vt + \frac{1}{2}v_0t$	
$= \frac{1}{2}(v_0 + at)t + \frac{1}{2}v_0t$	
$= v_0t + \frac{1}{2}at^2$	
From $\textcircled{1}$: $t = \frac{v - v_0}{a}$	✓
Substituting into above:	

$x - x_0 = v_0 t + \frac{1}{2} a t^2$ $= v_0 \left(\frac{v - v_0}{a} \right) + \frac{1}{2} a \left(\frac{v - v_0}{a} \right)^2$ $2(x - x_0)a = 2v_0(v - v_0) + (v - v_0)^2$ $2a\Delta x = 2v_0v - 2v_0^2 + v^2 + v_0^2 - 2v_0v$ $v^2 = v_0^2 + 2a(x - x_0)$	✓
$a = a_{avg} = \frac{v - v_0}{t - 0} \quad \text{①}$ $v_{avg} = \frac{x - x_0}{t - 0} \quad \text{②}$ $x = x_0 + v_{avg} t$ From ①: $t = \frac{v - v_0}{a}$ ③ Substituting ③ in ②: $\frac{(x - x_0)a}{v - v_0} = \frac{1}{2}(v + v_0)$ $2(x - x_0)a = (v + v_0)(v - v_0)$ $= v^2 - v_0^2$ $v^2 = v_0^2 + 2a(x - x_0)$	Shorter method

[3]

Question 7

9

A motorcyclist who is moving along an x axis directed toward the east has an acceleration given by $a = (6.1 - 1.2t) \text{ m.s}^{-2}$ for $0 \leq t \leq 6.0 \text{ s}$. At $t = 0 \text{ s}$, the velocity and position of the cyclist are 2.47 m.s^{-1} and 7.3 m respectively.

'n Motorfietsryer beweeg oos langs 'n x -as gerig en het 'n versnelling gegee deur $a = (6.1 - 1.2t) \text{ m.s}^{-2}$ vir $0 \leq t \leq 6.0 \text{ s}$. By $t = 0 \text{ s}$, is die die snelheid en posisie van die fietsryer $2,47 \text{ m.s}^{-1}$ en $7,3 \text{ m}$ onderskeidelik.

(b) What is the maximum speed achieved by the cyclist?

Wat is die maksimum spoed wat die motorfietsryer bereik?

[5]

$a = (6.1 - 1.2t)$ The velocity will be maximum when the first derivative of v ie. acceleration = 0 m/s^2.	
$a = (6.1 - 1.2t) = 0$ $t = \frac{6.1}{1.2} = 5.08 \approx 5.1 \text{ s}$ The velocity is obtained by the integrating the acceleration expression	✓ ✓

$a = (6.1 - 1.2t)$ $v = \int a dt = \int (6.1 - 1.2t) dt$ $= 6.1t - 0.6t^2 + c$ Since $v = 2.47$ m/s at $t = 0$ s, then $c = 2.47$ m/s Need to now determine the maximum speed at 5.1 s $v = [6.1(5.1) - 0.6(5.1)^2 + 2.47]$ $= 17.97 \text{ m.s}^{-1}$	
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(b) What total distance does the cyclist travel between $t = 0$ s and 6.0 s.

Wat is die totale afstand afgelê deur die ryer tussen $t = 0$ s en 6.0 s.

[4]

Total distance will be the area under the v - t curve (integral of the $v(t)$ function between 0 and 6 s. $v = 6.1t - 0.6t^2 + 2.47$ $x = \int v dt = \int_0^6 (6.1t - 0.6t^2 + 2.47) dt$ $= [3.05t^2 - 0.2t^3 + 2.47t]_0^6$ $= 81.4 \text{ m}$ You need to determine the x value (total distance) between the limits of $t = 0$ s and $t = 6$ s, since the starting point at $t = 0$ s is at $x = 7.3$ m.	
The following is not necessary, but could have been used to determine distance travelled. $x = \int v dt = \int_0^6 (6.1t - 0.6t^2 + 2.47) dt$ $= [3.05t^2 - 0.2t^3 + 2.47t + c]$ Since $x = 7.3$ m/s at $t = 0$ s, then $c = 7.3$ m $x_f = [3.05(6)^2 - 0.2(6)^3 + 2.47(6) + 7.3]$ $= 109.8 - 43.2 + 14.8 + 7.3$ $= 88.7 \text{ m}$ $\Delta x = x_f - x_i$ $= 88.7 - 7.3 = 81.4 \text{ m}$	

Question 8**8**

A small rocket, such as those used for meteorological measurements of the atmosphere, is launched vertically with an acceleration of 30 m/s^2 . It runs out of fuel after 30 s. What is the maximum altitude of the rocket?

'n Klein vuurpyl, soos dié wat gebruik word vir weerkundige metings van die atmosfeer, word vertikaal gelanseer met 'n versnelling van 30 m/s^2 . Die brandstof raak op na 30 s. Wat is die maksimum hoogte van die vuurpyl?

[8]

There are two stages to this problem. Firstly, when the rocket is firing and using fuel, secondly when the fuel is finished, the rocket is then in free-fall.

Phase 1 (before fuel runs out):

$$v_o = 0 \text{ m/s} \quad a_1 = +30 \text{ m/s}^2 \quad t = 30 \text{ s} \quad y_1 = ?$$

$$v_1 = ?$$

$$\begin{aligned} v_1 &= v_o + a_1 t & y_1 &= y_o + v_o t + \frac{1}{2} a_1 t^2 \\ &= (0 \text{ m} \cdot \text{s}^{-2})(30 \text{ s}) \quad \checkmark \checkmark & &= 0 \text{ m} + (0 \text{ m} \cdot \text{s}^{-1})(30 \text{ s}) + \frac{1}{2} (30 \text{ m} \cdot \text{s}^{-2})(30 \text{ s})^2 \quad \checkmark \checkmark \\ &= 900 \text{ m} \cdot \text{s}^{-1} & \Delta y_1 &= 13500 \text{ m} \end{aligned}$$

The rocket is moving upwards at a velocity of 900 m/s. It has not yet reached its maximum height.

Phase 2 takes care of the distance travelled till $v_y = 0 \text{ m/s}$.

Phase 2 (after fuel runs out):

$$v_1 = 900 \text{ m/s} \quad a_1 = -9.8 \text{ m/s}^2 \quad y_2 = ?$$

$$v_2 = 0 \text{ m/s} \quad \checkmark$$

$$\begin{aligned} v_2^2 &= v_1^2 + 2a_2 \Delta y \\ \Delta y &= \frac{v_2^2 - v_1^2}{2a_2} \\ &= \frac{(0 \text{ m} \cdot \text{s}^{-1})^2 - (900 \text{ m} \cdot \text{s}^{-1})^2}{2(-9.8 \text{ m} \cdot \text{s}^{-2})} \quad \checkmark \\ &= 41326 \text{ m} \end{aligned}$$

$$\begin{aligned} \Delta y &= \Delta y_1 + \Delta y_2 = 13.5 \times 10^3 \text{ m} + 41.3 \times 10^3 \text{ m} \quad \checkmark \checkmark \\ &= 54.8 \times 10^3 \text{ m} \end{aligned}$$

ENJOY THE WEEKEND / GENIET DIE NAWEEK